Computer Hardware Tasks

RAM and Storage: Comparison, Inspection and Reinstallation

Task 1: DDR3 vs DDR4 vs DDR5

	DDR3	DDR4	DDR5
Speed (typical)	800-2133 MT/s	2133-3200 MT/s (up to ~5000 with overclocking)	4800-8000+ MT/s
Voltage	1.5 V (1.35 V for DDR3L)	1.2 V	1.1 V
Pins (desktop)	240	288	288 (different notch position)
Key features	Baseline	Higher density, lower power	On-die ECC, power management on the module, two channels per stick
Typical use	Older PCs and laptops (~2007-2016), legacy systems	Most PCs and laptops from 2015-2022, budget builds	New high-end PCs, gaming, workstations, servers

None of the three types is compatible with another's slot.

Task 2: RAM Module Inspection and Reinstallation

Photo of Installed RAM Module

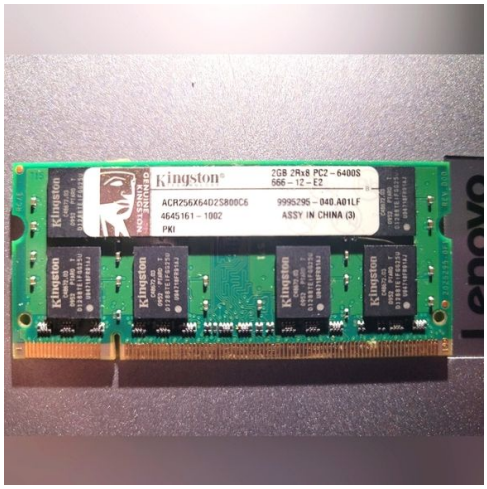


Figure 1: Kingston 2 GB DDR2-800 SO-DIMM removed from the laptop.

Module Identified from the Label

- **Brand:** Kingston
- **Form factor:** SO-DIMM (laptop memory)
- **Type and speed:** DDR2-800 (PC2-6400S)
- **Capacity:** 2 GB, 2Rx8 (dual rank)
- **Part number:** ACR256X64D2S800C6

Procedure

1. Shut down the laptop, disconnected the charger, and removed the battery.
2. Discharged static by touching bare metal.
3. Opened the access panel and located the RAM slot.
4. Photographed the module with its label visible.
5. Pushed the side clips outward, removed the module by its edges, and reinserted it at an angle, aligning the notch.
6. Pressed it down until the clips clicked, closed the panel, and powered on.

Verification

The system booted successfully, and [Task Manager / BIOS / System Information] reports [2 GB] of installed memory.

Observation

This is DDR2, an older generation than DDR3, DDR4 and DDR5. It runs at about 1.8 V, which is higher than all three newer types.

Task 3: HDD vs SSD vs NVMe

Three Key Differences

1. **Speed:** HDD is ~100-200 MB/s, SATA SSD ~550 MB/s, NVMe ~3,500-14,000 MB/s.
2. **Technology and interface:** HDDs use spinning platters (SATA). SSDs use flash with no moving parts (SATA). NVMe uses flash over PCIe, usually in an M.2 slot.
3. **Cost and durability:** HDDs are cheapest per GB but fragile and slow. SSDs cost more but are shock-resistant. NVMe is the most expensive but fastest.

Best-Fit Use for Each

- **HDD:** Photo, video and backup archives (Google Photos exports, Time Machine or File History backups).
- **SATA SSD:** Boot drive and everyday apps like Chrome, Office and Spotify. Startup feels instant.
- **NVMe:** Video editing (Premiere, CapCut on PC), large game loads, and big file transfers.

Task 4: Storage Drive Removal and Reinstallation

Photo of Drive and Connector



Figure 2: Samsung PM9A1 512 GB M.2 NVMe SSD in the laptop.

Drive Identified from the Label

- **Drive type:** SSD (flash storage, no moving parts)
- **Brand and series:** Samsung PM9A1
- **Model / part number:** MZ-VL25120 / MZVL2512HCJQ-00BL7 (Lenovo P/N SSS0L25196)
- **Capacity:** 512 GB
- **Form factor:** M.2 (2280 gum-stick card)
- **Connector / interface:** M.2 NVMe over PCIe Gen4 x4 (label reads "PCIe Gen4x4" and "PCIe NVMe")
- **Power rating:** 3.3 V, 2.9 A

Procedure

1. Shut down the laptop, disconnected the charger, and disabled or disconnected the internal battery as the device allows.
2. Touched bare metal to discharge static.
3. Removed the bottom cover and located the M.2 drive slot.
4. Photographed the drive and its label while installed.
5. Removed the retaining screw; the drive lifted at an angle, and I pulled it out of the slot by its edges.
6. Reinserted the drive at an angle into the M.2 slot, pressed it flat, and refastened the screw.
7. Replaced the cover, reconnected power, and booted the laptop.

Verification

The device booted successfully, and [Disk Management / Task Manager / BIOS] recognizes the drive as a [512 GB / about 476 GB] NVMe SSD.

Observation

The drive uses an M.2 NVMe connector rather than SATA, so it communicates over PCIe and is much faster than a SATA SSD or HDD (see Task 3). The M-key notch on the connector edge keeps it from being inserted incorrectly.